



Álvaro Buendía Gallego

Personal Details

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Introduction

I am a theoretical and computational physicist, specialized in the study of plasmonic and excitonic optical nanostructures. I use and develop semi-analytical methods and models to study, predict and tune the topological properties of periodic and finite lattices of nanoparticles.

Professional experience

- 2025-2026 **Postdoctoral Research Fellow at Iberian International Nanotechnology Laboratory (INL), Braga, Portugal**, IP: *Nuno Peres*.
- 2020-2024 **Predocutorial researcher (PRE2019-090689) at Instituto de Estructura de la Materia (IEM-CSIC), Madrid Spain**, IP: *José Antonio Sánchez Gil, Vincenzo Giannini*.
- 2018-2020 **Predocutorial researcher (Garantía Juvenil) at Condensed Matter Department at Universidad Autónoma de Madrid**, IP: *Elsa Prada*.

Research stays

- May-August 2023 **Department of Physics and Astronomy, University of Exeter (UK)**, IP: *Marie Rider, Bill Barnes*.
- July 2022 **Instituto de Telecomunicações, Lisboa (Portugal)**, IP: *Paloma A. Huidobro*.

Education

- 2020-2024 **PhD in Condensed Matter, Nanoscience and Biophysics**, Universidad Autónoma de Madrid (UAM), PhD Thesis, Topological Nanophotonics: Symmetry Protected States of Arrays of Plasmonic Nanoparticles..
Cum laude and International mentions
- 2017-2018 **Master's degree in Quantum Science and Technology**, Euskal Herriko Unibertsitatea/Universidad del País Vasco.
- 2013-2017 **Bachelor's degree in Physics**, Universidad Autónoma de Madrid.

Conferences

- 2026 **Iberian-German WE-Heraeus Seminar 2026, INL, Braga, Portugal**.

- 2025 **European Optical Society Annual Meeting**, TU-Delft, Delft, Netherlands.
- 2024 **Complex Nanophotonics Science Camp**, Cumberland Lodge, Windsor, UK.
- 2024 **Reunión Nacional de Óptica RNO 2024**, Universidad de Murcia, Murcia, Spain.
- 2023 **NANOP 2023**, Universidad Pompeu Fabra, Barcelona, Spain.
- 2023 **London Plasmonics Forum**, King's College, London, UK.
- 2022 **TERAMETANANO 2022**, Praiamar Hotel, Natal, Brazil.
- 2022 **Topological Photonics 2022 (TopoPhoto2022)**, DIPIC, San Sebastián, Spain.
- 2022 **NanoSpain Conf 2022**, Escuela Superior de Técnicos Industriales (ETSII), Madrid, Spain.
- 2021 **Topology meets Quantum Optics (2021TMQO)**, Online.

Presentations

Oral

- 2024 **"Long-range molecular energy transfer mediated by strong coupling to plasmonic edge states" (Invited Talk)**, Serrano Mini Workshop Series on Nanophotonics, Quantum Technologies and Information.
- 2024 **"Long-range molecular energy transfer mediated by strong coupling to plasmonic topological edge states"**, RNO 2024.
- 2022 **"Minimizing effect of long-range interactions in topological plasmonic arrays"**, TERAMETANANO 2022.
- 2022 **"Exploiting orientation of anisotropic nanoparticles in topological plasmonic arrays"**, NanoSpain2022.

Poster

- 2026 **"Multiple topological edge states in a Su-Schrieffer-Heeger chain of resonant core-shell nanoparticles"**, Iberian-German WE Heraeus Seminar 2026.
- 2025 **"Long-range molecular energy transfer mediated by strong coupling to plasmonic topological edge states"**, EOSAM 2025.
- 2024 **"Long-range molecular energy transfer mediated by strong coupling to plasmonic topological edge states"**, Complex Nanophotonics Summer Camp.
- 2023 **"Strong coupling between molecules and topological edge states of arrays of plasmonic nanoparticles"**, NANOP 2023.
- 2022 **"Exploiting orientation of anisotropic nanoparticles in topological plasmonic arrays"**, TopoPhoto2022.

Publications

- 2026 **Far-field radiation of bulk, edge and corner eigenmodes from a finite two-dimensional Su-Schrieffer-Heeger plasmonic lattice**, Á. Buendía (Correspondant Author), J. L. Pura, J. A. Sánchez-Gil, V. Giannini, Physical Review B (Accepted, <https://journals.aps.org/prb/accepted/10.1103/54vz-h71c>).
- 2026 **Coupled electric dipole model for a Su-Schrieffer-Heeger chain of optically resonant coreshell nanoparticles**, Á. Buendía (Correspondant Author), Nuno Peres, J. Appl. Phys. 139, 063110.
- 2024 **Long-range molecular energy transfer mediated by strong coupling to plasmonic topological edge states**, Á. Buendía (Correspondant Author), J. A. Sánchez-Gil, V. Giannini, W.L. Barnes and M. S. Rider, Nanophotonics. <https://doi.org/10.1515/nanoph-2024-0077>

- 2023 **Exploiting Oriented Field Projectors to Open Topological Gaps in Plasmonic Nanoparticle Arrays**, Á. Buendía (Correspondant Author), J. A. Sánchez-Gil, V. Giannini, ACS Photonics 2023, 10, 2, 464–474, <https://doi.org/10.1021/acsp Photonics.2c01526>.
- 2022 **Optical Mie Scattering by DNA-Assembled Three-Dimensional Gold Nanoparticle Superlattice Crystals**, HJ Singh, D Misatziou, C Wheeler, Á Buendía, V Giannini, JA Sánchez-Gil, Martinus HV Werts, Tom Brown, Afaf H El-Sagheer, Antonios G Kanaras, Otto L Muskens, ACS Applied Optical Materials 1 (1), 69-77, <https://doi.org/10.1021/acsaom.2c00008>.
- 2022 **Advances and Prospects in Topological Nanoparticle Photonics**, M S. Rider, Á. Buendía, D. R. Abujetas, P. A. Huidobro, J. A. Sánchez-Gil, V. Giannini, ACS Photonics 2022, 9, 5, 1483–1499. <https://doi.org/10.1021/acsp Photonics.1c01874>
- 2020 **Tailoring pi-conjugation and vibrational modes to steer on-surface synthesis of pentalene-bridged ladder polymers**, Bruno De la Torre, Adam Matěj, Ana Sánchez-Grande, Borja Cirera, Benjamin Mallada, Eider Rodríguez-Sánchez, José Santos, Jesús I Mendieta-Moreno, Shayan Edalatmanesh, Koen Lauwaet, Michal Otyepka, Miroslav Medved', Álvaro Buendía, Rodolfo Miranda, Nazario Martín, Pavel Jelínek, David Ěcija, Nature communications 11 (1), 4567.
- 2019 **S-matrix poles symmetries for non-Hermitian scattering Hamiltonians**. M.A.Simon, A. Buendía, A. Kiely, A. Mostafazadeh, J.G. Muga, Physical Review A 99 (5), 052110 (2019),
- 2018 **Symmetries and Invariants for Non-Hermitian Hamiltonians**. M. A. Simón, A. Buendía and J.G. Muga, Mathematics, 6, 111 (2018)

Teaching

- 2025 **Tutorial of coupled electric dipole formalism for chains of nanoparticles**, Nano-ótica, Master Física, UMINHO, Braga, Portugal.

Courses

- 2022 **How to prepare scientific presentations**, *Spanish Research Council, Madrid, Spain*.
- 2019 **Summer school on Quantum Technologies**, *Universidad Internacional Méndez Pelayo (UIMP), Santander, Spain*.

Awards

- 2025 **CSIC Relevant Thesis Award to "Topological Nanophotonics: Symmetry protected states of arrays of plasmonic nanoparticles"**.
- 2022 **Shortlisted story and People's Prize at Quantum Shorts scientific literary contest**.
- 2019 **Honorable mention at Qiskit Hackathon Madrid to the project CryptoQ**.
- 2013 **Spanish Army National Award to the project "Semaphore Line Communication"**.

Reviewing

- 2025 **Review of a paper for Science Advances**.
- 2024 **Review of a paper for Optics Express**.
- 2023 **Review of a paper for Advanced Optical Materials**.

Miscellaneous

- 2025-2026 **Member of Spanish Optical Society (SEDOPTICA)**.

- 2025 **Workshop, European Researchers Night.**
- 2025 **Workshop, Nanoexplorers, INL Open Day.**
- 2023-2024 **Organizer PhDay Serrano, *PhDay*.**
- 2024 **Mentee at CAMINO CSIC mentoring program.**
- 2023 **Board member of CSIC PhD Student Network and IOPTICA Student Chapter.**
- 2022-2024 **Workshop "Discovering light" at Science Week, high and middle schools.**

— Languages

Spanish	Native.	Portuguese	Basic.
English	C1.	French	Basic.

— Programming languages and software

C++, Python, Julia, Matlab, Mathematica, Comsol